

Scrapers Documentation

IT-Hell Scrapers is a data collection project designed to concurrently scrape job listings from popular Polish IT job boards (pracuj.pl, theprotocol.it). It utilizes asynchronous requests and browser impersonation to reliably extract job offers, parses them into structured data models, and pushes the results into a Redis Stream for further processing.

Getting started

1. Environment Setup:

- Ensure you have Python 3.8+ installed.
- Install the uv package manager if you haven't installed it yet.
- Ensure you have configured Redis correctly.

2. Install dependencies:

- Run `uv sync` to install the required dependencies from `pyproject.toml`.

3. Setup environment variables:

- Create a `.env` file in the root of the project with the following content (adjust values as needed):

```
REDIS_HOST=  
REDIS_PORT=  
REDIS_DB=  
REDIS_PASSWORD=  
REDIS_STREAM=  
REDIS_GROUP=  
REDIS_CONSUMER=
```

4. Configure scrapers (optional):

- Edit `app-config.yaml` to adjust scraper settings.

5. Run scrapers:

1. Run all scrapers:
`uv run main.py`
2. Run specific scraper (pracuj_pl, theprotocol_it):
`uv run main.py -s scraper_name`

Architecture and Data Flow

1. **Runner (main.py):** Acts as the asynchronous entry point. It instantiates the chosen scraper classes and runs their main loops concurrently.
2. **Scraping Engine:**
 - Uses `curl_cffi` for HTTP requests, applying TLS fingerprinting (browser impersonation) to bypass basic anti-bot defenses.
 - Paginates through listings, fetches job HTML pages, and places items into an `asyncio.Queue`.
3. **Data Parsing & Validation:**
 - The raw HTML is parsed using helper functions (e.g., extracting JSON data from `<script>` tags).
 - The data is mapped to structured and validated Pydantic models (`JobOffer`).
4. **Redis Integration:**
 - A dedicated asynchronous worker consumes the enriched `JobOffer` items from a queue.
 - Serializes the items to JSON and pushes them to a Redis Stream (`REDIS_STREAM`) using the `XADD` command.

Data flow diagram



File Structure

```
scrapers/
├── .env          # Environment variables
├── main.py       # Entry point for running scrapers
├── pyproject.toml # Project dependencies managed by uv
├── README.md     # Documentation
├── app-config.yaml # Configuration file for scraper settings
├── logs/         # Application log files
├── src/
│   ├── core/     # Core modules (e.g., Redis client connection)
│   ├── pracuj_pl/ # Scraper implementation and data extraction for pracuj.pl
│   ├── theprotocol_it/ # Scraper implementation and data extraction for theprotocol.it
│   ├── helpers.py # HTML to JSON parsing utilities
│   └── schemas.py # Pydantic data models for validation
└── tests/        # Pytest test suite and mock HTML data
```